

Introduction to Biology

Biology is the scientific study of life and living organisms. Living organisms include animals, plants, fungi, bacteria, and many other forms of life.

Cells are the basic structural and functional units of living organisms. Some organisms consist of a single cell, while others are made of many cells.

The major branches of biology include botany, zoology, microbiology, ecology, genetics, and molecular biology.

The Cell

A typical cell contains several important structures. The cell membrane controls what enters and leaves the cell. The nucleus contains genetic material and helps control cellular activities.

Mitochondria are organelles involved in producing energy for the cell. In plant cells, chloroplasts carry out photosynthesis using light energy.

Cells can be broadly classified as prokaryotic or eukaryotic. Prokaryotic cells do not have a membrane-bound nucleus, whereas eukaryotic cells have a membrane-bound nucleus.

Photosynthesis

Photosynthesis is the process by which green plants use light energy to make chemical energy from carbon dioxide and water.

Chlorophyll is the green pigment that absorbs light energy. Photosynthesis primarily takes place in chloroplasts within plant cells.

Oxygen is released as a by-product of photosynthesis. The process is important because it provides food for plants and contributes oxygen to the atmosphere.